

NORD TSS — Coding-Style and Authorship Analysis

Based on the recovered sources in `E:\Dev\Ronny\TSS\Sources\` .

[VERIFIED] = read directly from the files. **[ASSUMPTION]** = stylistic inference, clearly argued but not provable from the archive alone.

1. The hands in the code

Hand 1 — Bo Lewendal: the base system

[VERIFIED] TSS1 line 1: `% NORD TIMESHARING SYSTEM BY BO LEWENDAL .`

The bulk of all five parts (scheduler, paging, file system, monitor calls, command processor, TTY handling) is one uniform, very recognizable style:

- **Dense multi-statement lines** — 3–6 instructions per line joined with `; :`
`LEV6, SAT NTTY; COPY CM2 DX ST ...` the whole kernel reads like this.
- **Uniform terse routine headers:** `%NAME / one-line purpose / register contract ( %T = USER NUMBER , %A = DEVICE ID ) / %RETURN` — nothing more.
- **A systematic local-symbol idiom:** work cells after the code
`( TCL, 0 / TCT, 0 ), then )FILL and )PCL <name>` to close the scope and free the names for reuse in the next routine. This `)KILL/)`PCL scoping runs through all five files — one mind, one discipline.
- **ALGOL-ish pseudo-code comments** for the hard algorithms, e.g. the page allocator (TSS1 lines 2917/3214): `% FOR J:=0 TO NDPGS-1 DO;` and stepwise `%*`-prefixed narration (TSS1 3029–3040). **[ASSUMPTION]** the habits of someone at home in ALGOL-style structured thinking, writing assembler.
- **Assembler metaprogramming:** the `GOVER/OVERL/OVERX` overlay macros (TSS3 head) that move assembled code and write it to disk *during assembly* — the most sophisticated MAC usage in the archive.
- Short reused labels (`L1 , L2 , MS1 ...`), message strings `'...$\ ' .`

Hand 2 — NJL = Nils Jakob Langeland: the 1973 NORD-10 device drivers

[VERIFIED] two signed items, both spring 1973:

- XDRUM — TSS1 3730–3874: *"D R U M T R A N S F E R R O U T I N E ... THIS IS 1. APPROXIMATION TO A NORD-10 VERSION — NJL 17/4/73"*
- IOXLIB — MINIT lines 592–596: *"STANDARD INPUT/OUTPUT FOR NORD-10 ... %>=...* IOXLIB ...=% — ORIGINATOR: NJL 28/5/73"**

The style could not be more different from Lewendal's:

- **One instruction per line**, operands in tab columns, an aligned % comment on nearly every line.
- **Letter-spaced banner titles** (% D R U M T R A N S F E R ...) and a formal interface contract in fixed sections: CALLING SEQUENCE / CALLED WITH / RETURN INFORMATION — with the standard multi-exit convention (error exit / busy exit / finished exit) and the tell-tale closing line
% B IS NOT USED BY THE ROUTINE .
- Device register maps declared as symbol tables with per-register comments (RCX= 0 %READ CORE ADDRESS ...),)KILL -ed again after the routine.
- Dates in dd/mm/yy.

[ASSUMPTION, strong] The MAGTP mag-tape driver (TSS3 455–560) is the same hand or the same house template: identical banner, identical section headings, identical % B IS NOT USED BY THE ROUTINE line, identical
% HDEV,B MUST HOLD THE HARDWARE DEV. NUMBER remark — but it is the NORD-1 ("NN10 , IOT) variant, so this template predates the N10 port. This is the documentation shape later seen in SINTRAN driver sources — i.e. Norsk Data house driver style. NJL is **Nils Jakob Langeland** (identification supplied by Ronny, 2026-07-19 — not stated in the archive itself, which only carries the initials).

Hand 3 — the N10 in-line patches (author uncertain)

The ~120 "N10 / "NN10 conditional blocks **[VERIFIED count: 97+104 in TSS1, 13+12 in TSS2, 5+10 in TSS3, 3+3 in TSS4, 2+2 in TSS5]** are mostly surgical 1–5 line substitutions written in **Lewendal's compact style**, not NJL's driver style — e.g. the whole N10 INIT , the IRW / IRR context patches, the VERDR Versatec driver. **[ASSUMPTION]** either Lewendal did the core N10 port himself, or the porter deliberately matched his style; the archive cannot distinguish. The XDRUM comment "1. approximation to a NORD-10 version" shows the N10 port was in progress in April 1973.

Hand 4 — imported table

[VERIFIED] TSS1 1685: % CARD CODE CONVERSION TABLE – % REVISED 1971.08.04 — a yyyy.mm.dd date, unlike NJL's dd/mm/yy; likely imported from an existing

source [ASSUMPTION]. Oldest date in the archive.

Hand 5 — "CVS": the modern restorer

[VERIFIED] % Added by CVS lines exist only in ASSYSA/ASSYSB/MINIT/TDUMP:
symbol patches for opcodes MAC no longer had built in, and the notes
STR has been substituted by XTR , LSS has been substituted by XSS . No code.

Version identity

[VERIFIED] TSS5 1085/1092: startup banners
'\$NORD TSS VERSION 3.0A IS UP\$\ ' and ...3.0B... — this archive is
TSS version 3.0, A = production, B = DEBUG build. TSS4 1724 prompts
ID[YEAR (E.G. 1973):] — consistent with a 1973 system.

2. What is specifically written for the NORD-10 ("N10 code)

The NORD-1 ("NN10) is the original target; every N10 block replaces a
NORD-1 mechanism with the NORD-10 equivalent. Complete inventory of the
N10-specific machinery [VERIFIED locations]:

Area	NORD-1 way ("NN10)	NORD-10 way ("N10)	Where
Interrupt entry	Memory vector blocks (20/61; RBLOK; ...) + software SAVE / UNSAV of registers to memory	Hardware register levels; program level-P registers with IRW nn DP ; IDENT PL11 vectored device identification	TSS1 178–206, 236–245, 1797– 1811
Interrupt init	IOT 6 , poke vectors via STA I (61...	POF , TRR PIE , IRW , TRR IIE , ION	TSS1 219–291 (INIT fully rewritten)
Interrupt enable/disable	INTDS=150440 / INTEN=150540 (device words)	INTDS=IOF , INTEN=ION (CPU instructions)	TSS1 103–106
Error accounting	—	NER10/NER11/NER12/NIOX/SER23/SER10/SER12 false-interrupt and IOX-timeout counters + restart entry 20/ JMP I *+1; 301	TSS1 193–206
Paging / memory	4K/16K MPR protect register (TRR MPR)	IPGTB "INITIALIZE PAGING SYSTEM": loop TRR PCR writing paging-control registers,	TSS1 236–291,

Area	NORD-1 way ("NN10)	NORD-10 way ("N10)	Where
protection		LDA (1136; TRR IIE , POF/PON	305+
Context switch	RBLOK register block in memory, STT I (RBLOK	Level-1 hardware register block: IRW 10 DP , IRR 10 DP , LRB/SRB 10	TSS2 380–384, 962–965, 993–995, 1734– 1759
Divide	Software DIV routine (~20 lines, PROGM/ENTER)	Hardware RDIV : DIV, SAD ZIN SHR 20; RDIV ST; EXIT — one line	TSS2 2165– 2179
Real-time clock	IOT PIN 6 / IOT 6 (OLA variant for "special RTC")	IOX 13 with control words 122001 (on) / 20001 (off), preset LDA (310; IOX 11	TSS2 1796– 1814, TSS1 288–289
TTY / paper tape I/O	IOT with SNI/ACT/SKA/PIN bit combinations	IOX register model: dev+0 data, +2 status (BSKP ... 30 DA ready bits), +3 control; TTY at 300, reader at 400	MINIT IOXLIB 590–655, TSS1 TTY blocks (1580– 1660 area)
Swapping drum	<i>(does not exist on NORD-1 config — disk only, TRSFR=TRXX)</i>	XDRUM driver, device IOX 540 +0..7, plus TRSFR drum/disk router	TSS1 3711– 3874
CDC disk	IOT -style channel (DISC=DCHN+44 etc.)	N10 register set LCA=DCHN+1, LBA=+3, LMR=+5, RST=+4, LWC=+7	TSS1 3341– 3600 "CDC N10 blocks, TDUMP 51–56

Area	NORD-1 way ("NN10)	NORD-10 way ("N10)	Where
Versatec printer	—	VERDR driver ("N10 VERSA mark), IOX DLP+1/3/5 , hangs off LEV11 DMA ident	TSS1 1813– 1837
Mag tape	Full MAGTP driver (Kennedy or HP variants, IOT)	No N10 driver present — the driver is "NN10 -only	TSS3 455+
Plotter output	IOT ACT SKA PLOT busy-wait	plain jump — plotter I/O dropped/deferred	TSS2 1351– 1355
Enable mask	ENABL, 65357 + SVEEN, 3717	ENABL, 77157	TSS1 296–301

Reading of the table **[ASSUMPTION]**: the N10 port was done in 1973 by rewriting exactly the machine-dependent layer — interrupts, context, paging, clock, and each device driver — while every line of scheduler/file-system/command logic is shared between both targets via the mark system. Mag tape and plotter were not (yet) ported; the drum and Versatec exist *only* for the NORD-10 — the drum being the point of the port (real swapping hardware), and XDRUM's own header admits it is a first approximation still being brought up.

3. Quick fingerprint guide

Trait	Lewendal (base)	Langeland (drivers)
Line density	3–6 instructions per ; -line	one instruction per line
Comments	header contract only; pseudo-code for algorithms	aligned % on almost every line
Banners	%NAME	% D R U M ... letter-spaced
Interface doc	%T = ... one-liners	CALLING SEQUENCE / CALLED WITH / RETURN INFORMATION sections
Exits	RET/EXIT, skip-returns	formalized error/busy/finished multi-exit
Dates	none	dd/mm/yy (17/4/73, 28/5/73)